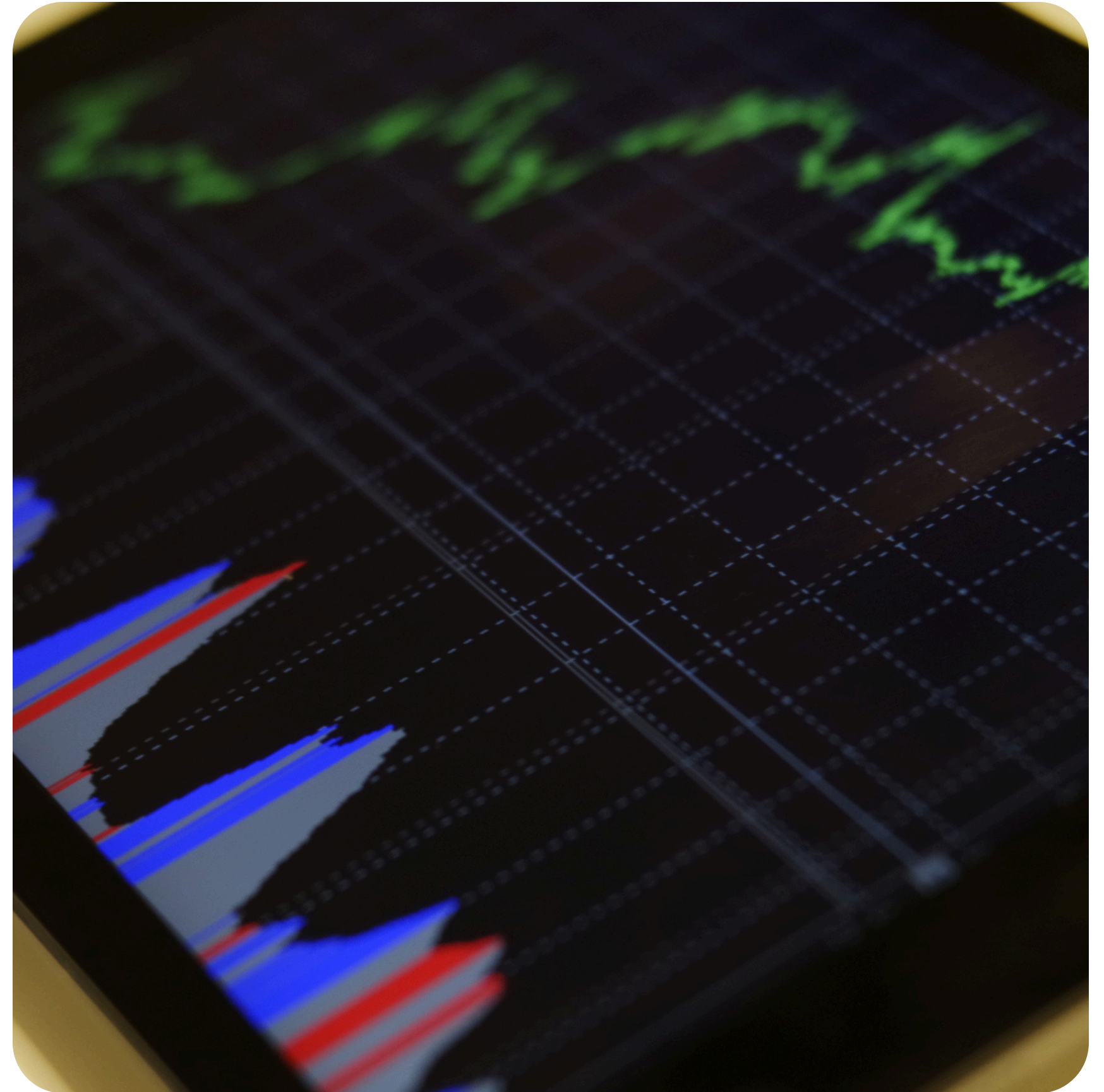


RETAIL : ↗
CUSTOMER PURCHASE
BEHAVIOR
& SALES PERFORMANCE
ANALYSIS



GROUP MEMBERS

Nada Moustafa

Tasneem

Abdallah Mohamed

Retage wael mohamed

Mariam Mohamed Mohamed Elsayed



WHAT IS PROBLEM ?

The company has two years of sales data (2009–2011), but lacks clear visibility into: which products sell best, who the key customers are, and how sales vary across months and countries. Without this analysis, marketing and inventory decisions are made based on guesswork, not data.

Project Objective

The main objective of this project is to analyze retail sales data and transform raw data into meaningful business insights that support better business decisions.

Our objectives include:

- **Analyze product performance.**
- **Understand customer purchasing behavior.**
- **Identify sales trends over time.**
- **Compare sales performance across countries.**
- **Support data-driven decision-making.**



KPI

- Total Revenue
- Total Orders
- Total Customers
- Total Quantity Sold
- Average Order Value

CLEANING

We Handled :-

- 1 Missing values
- 2 Zero or negative values
- 3 Duplicate values
- 4 DataFrame
- 5 Cancelled orders

DATA SET OVERVIEW

Number of rows

1,029,405

Number of columns

7 column

Number of customers

6000+

Number of countries

43

Number of products

5000+

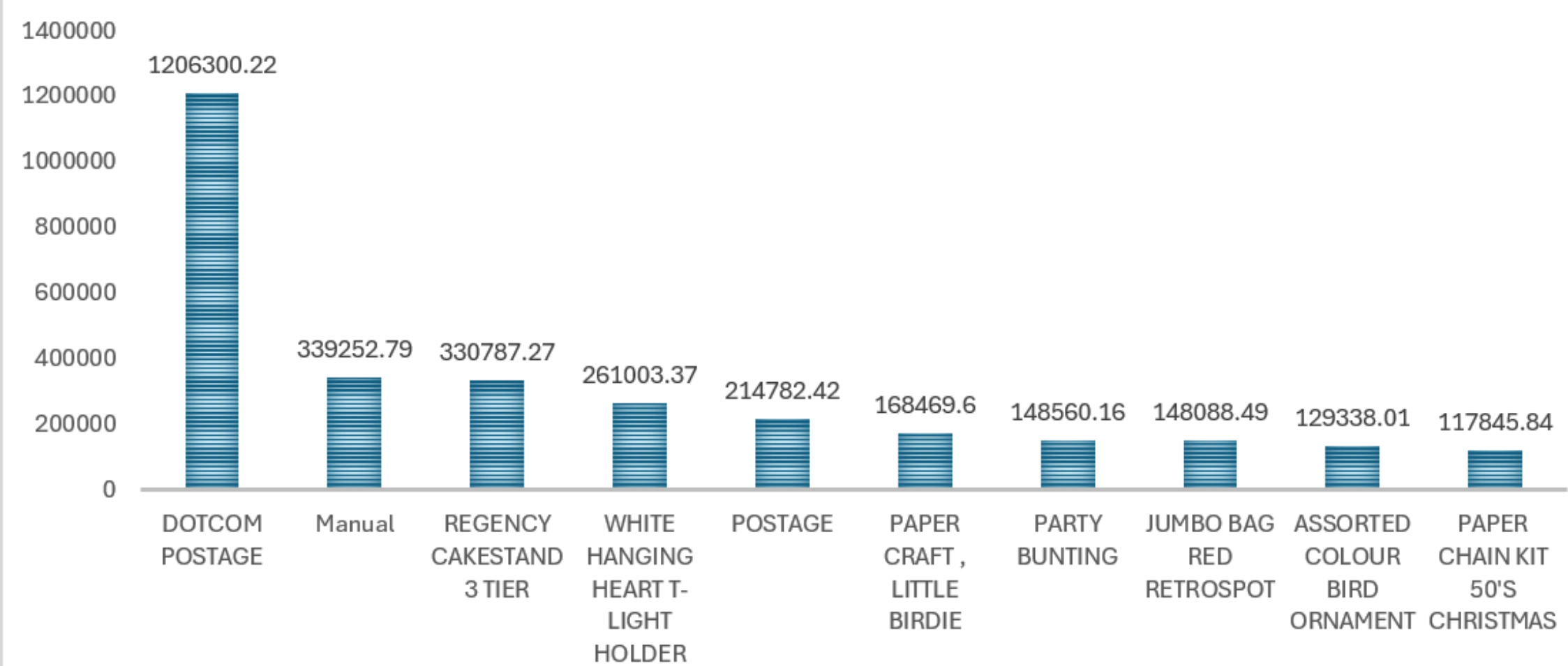


TOP BUSINESS QUESTIONS

- 1 What are the top 10 products by revenue and by quantity sold?
- 2 How does monthly revenue trend over the two years?
- 3 Which countries generate the most revenue, and how concentrated is it?
- 4 Which customers order most frequently, and what is their average order value?

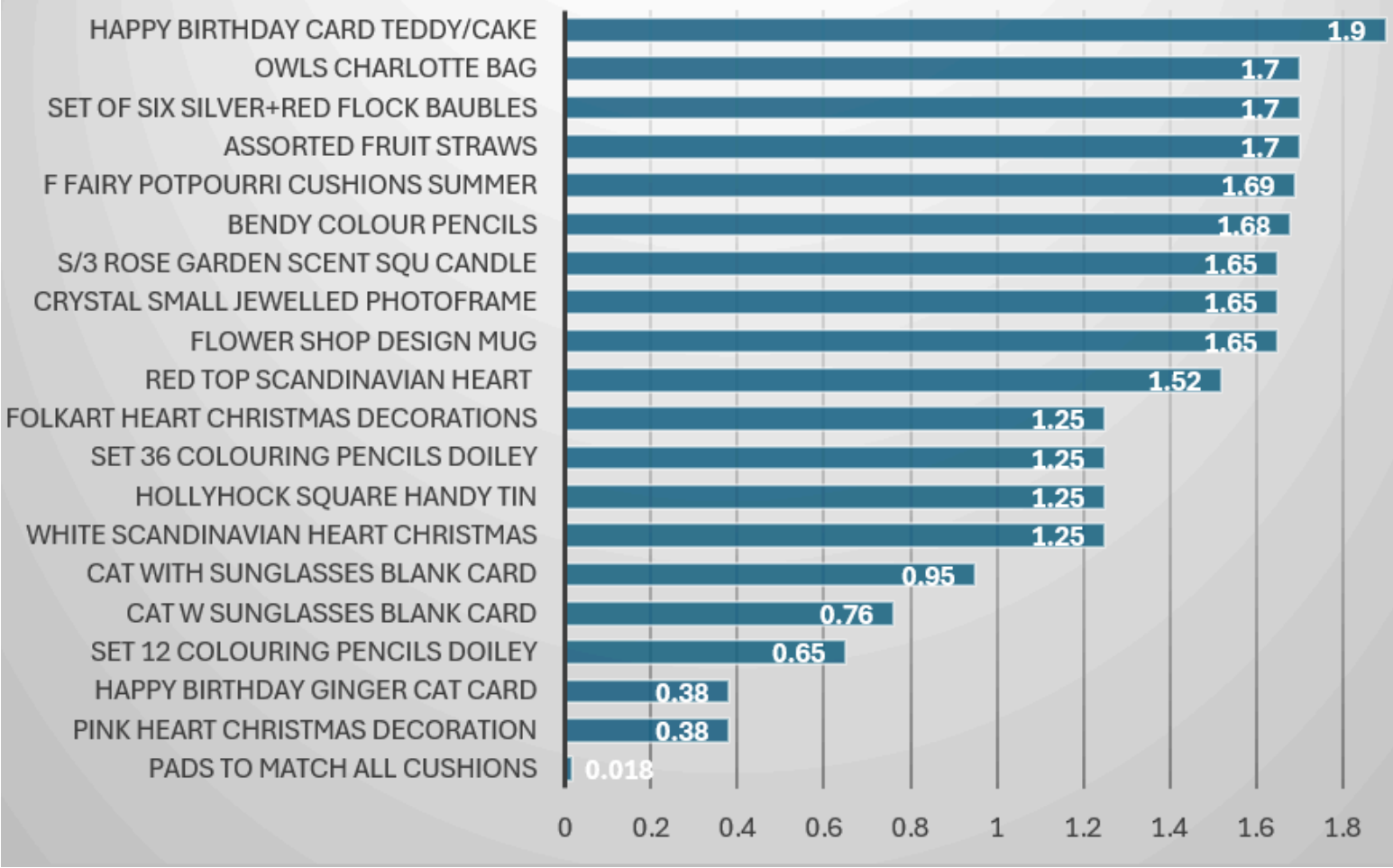
PRODUCTS

TOP 10 PRODUCTS BY REVENUE



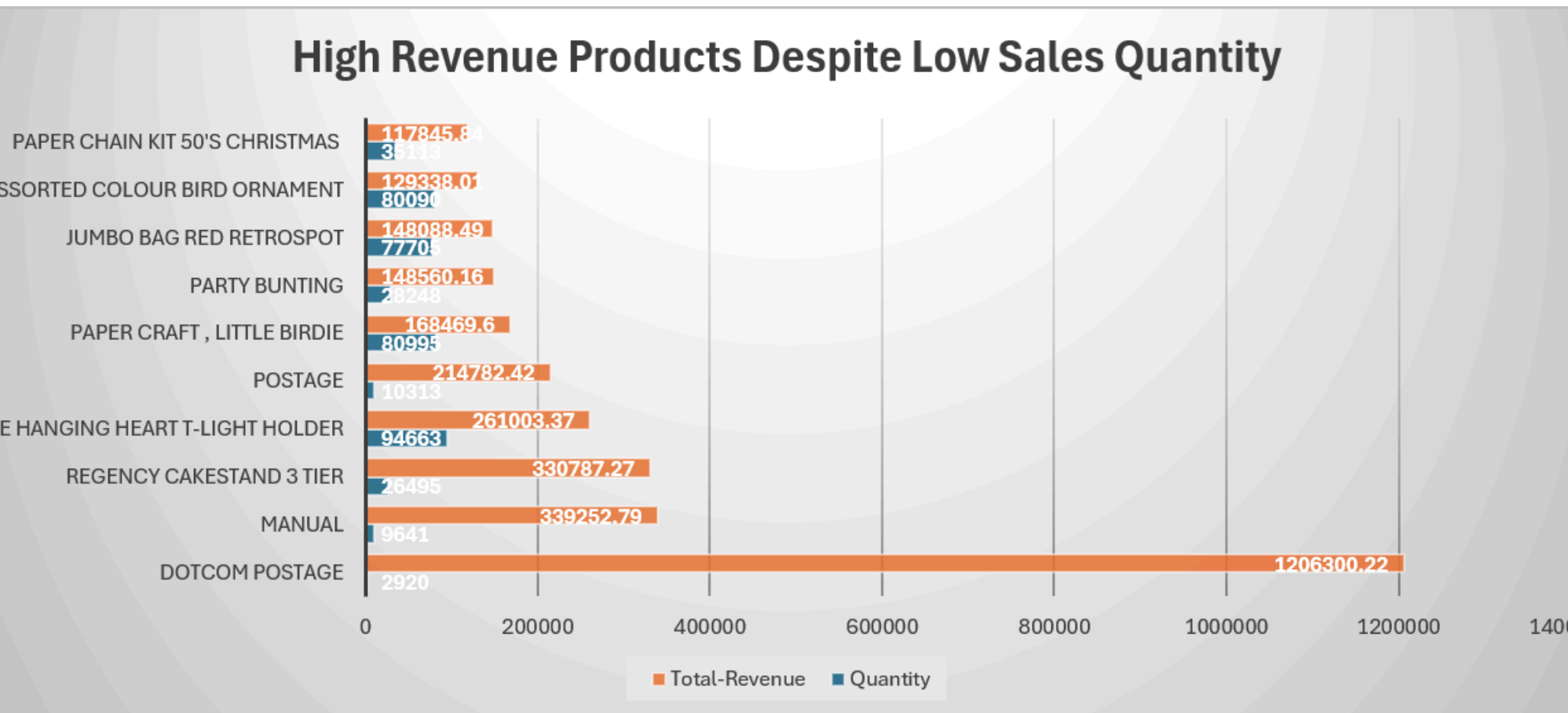
DOTCOM POSTAGE generated the highest revenue among all products

Lowest Revenue Products (Bottom 20)

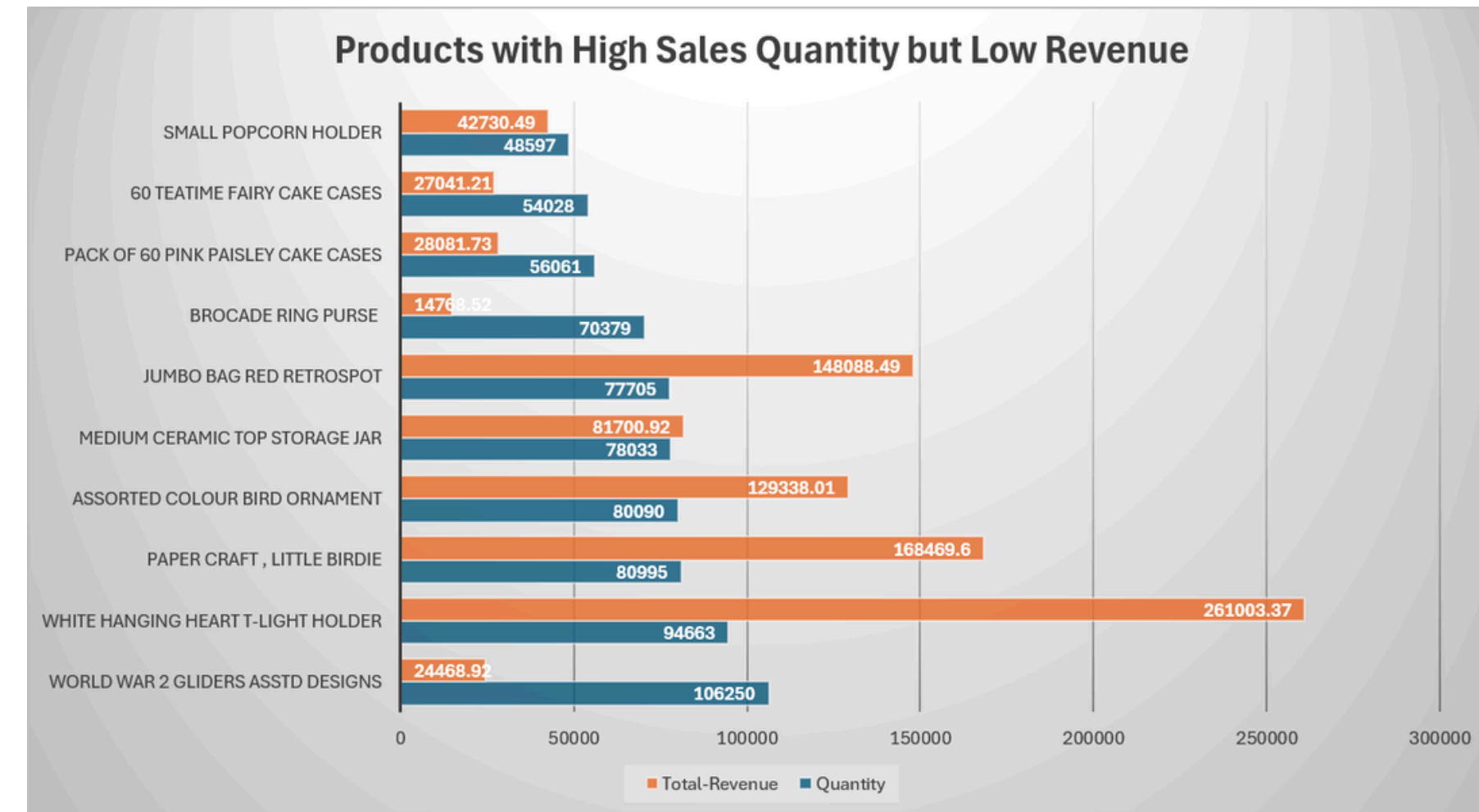


These products generated the lowest revenue and may require further evaluation

PRODUCTS



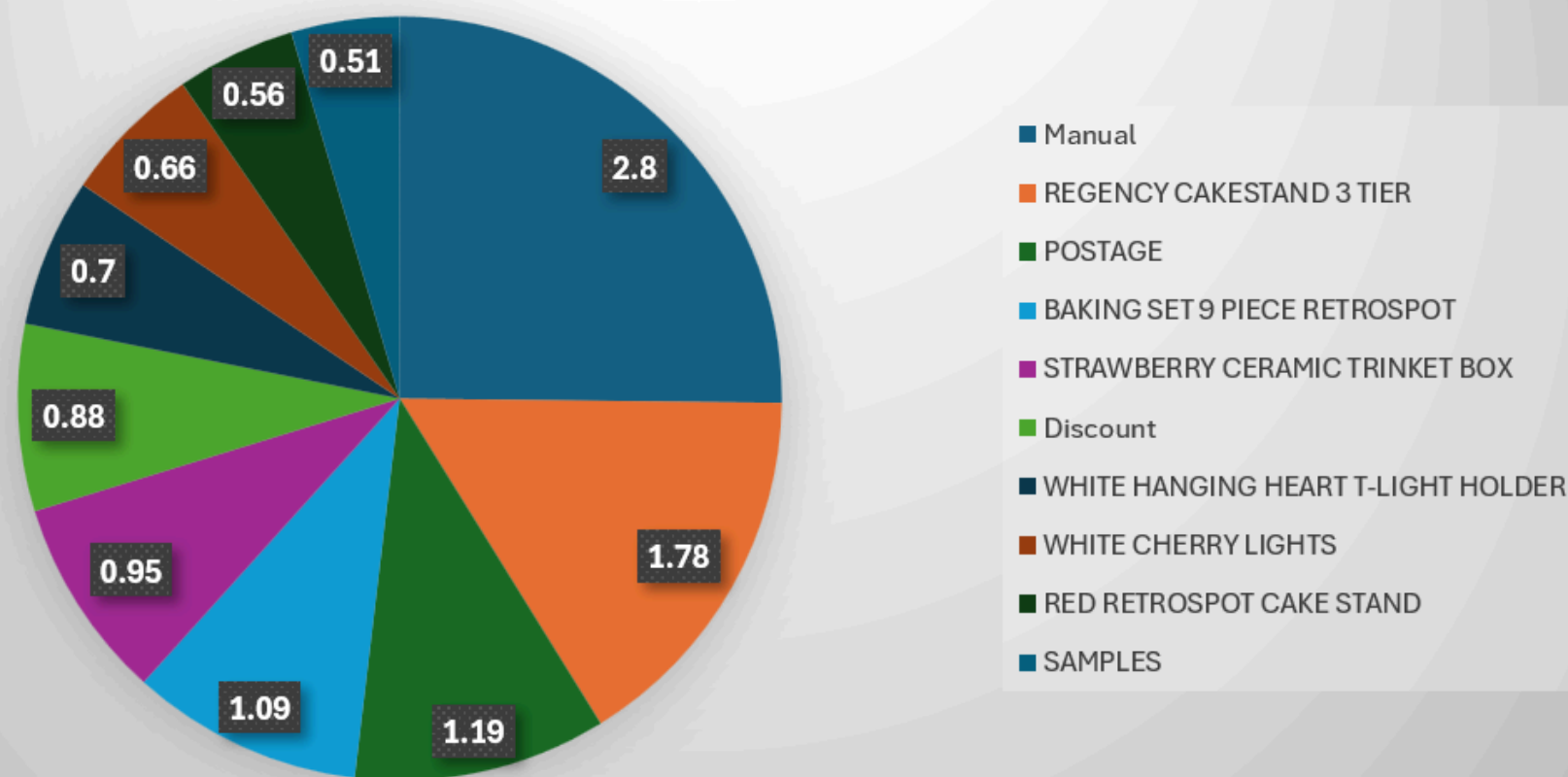
High-value products generate strong revenue despite low sales volume



High sales volume does not always mean high revenue.

PRODUCTS

Most Frequently Canceled Products



These products have the highest cancellation rates

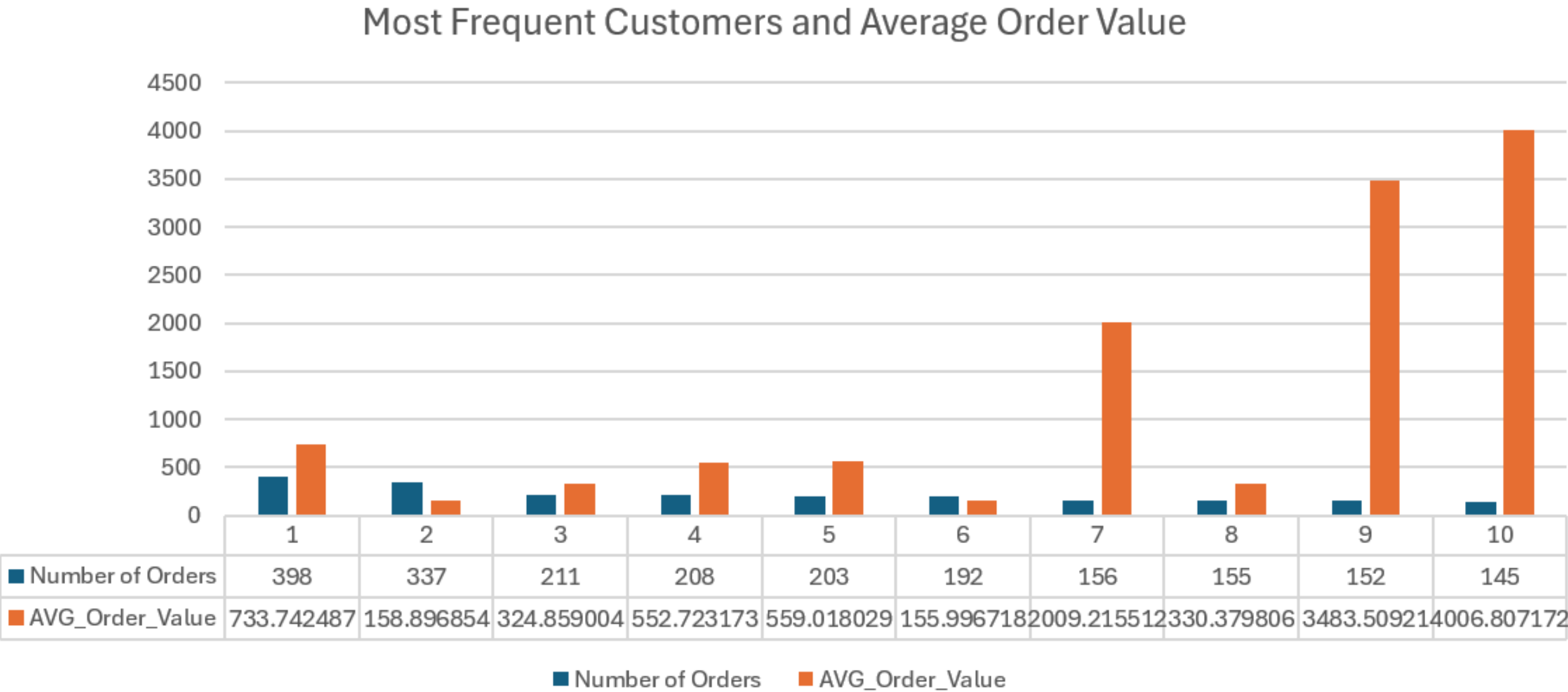
Top 10 Product Pairs Purchased Together



Customers often buy these products together

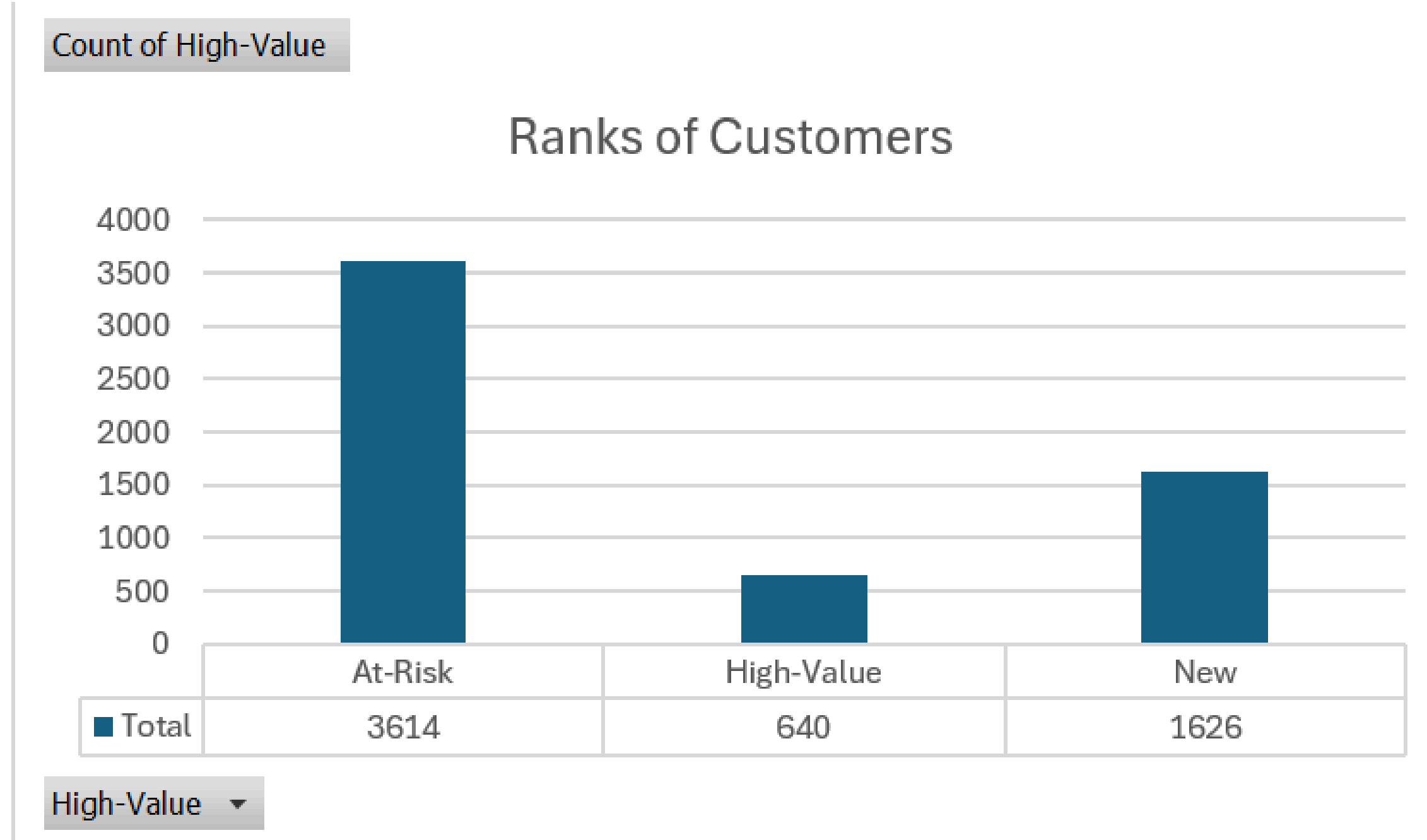
CUSTOMERS

1	14911
2	12748
3	17841
4	15311
5	13089
6	14606
7	14156
8	17850
9	14646
10	18102



Customer order frequency and average order value vary among customers

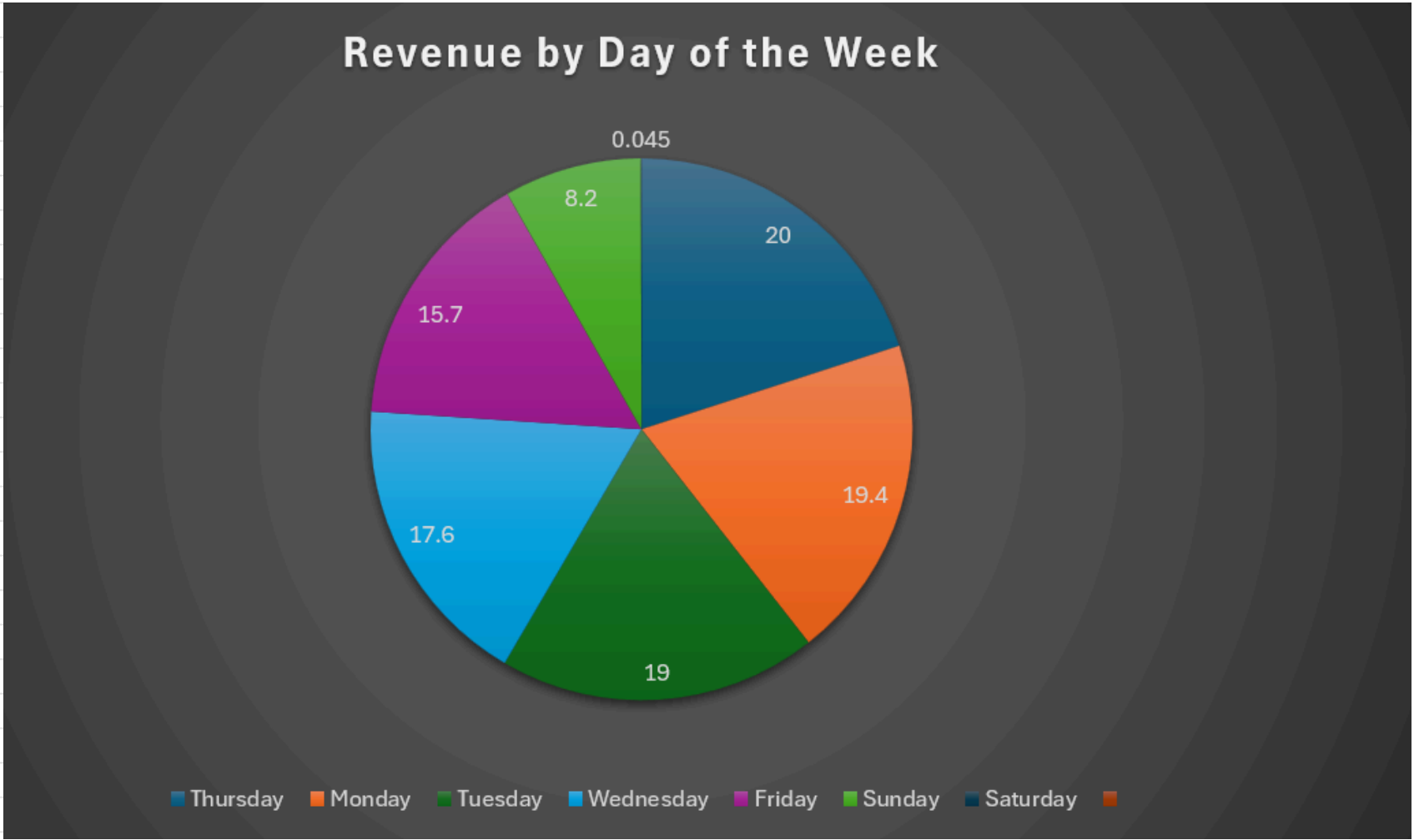
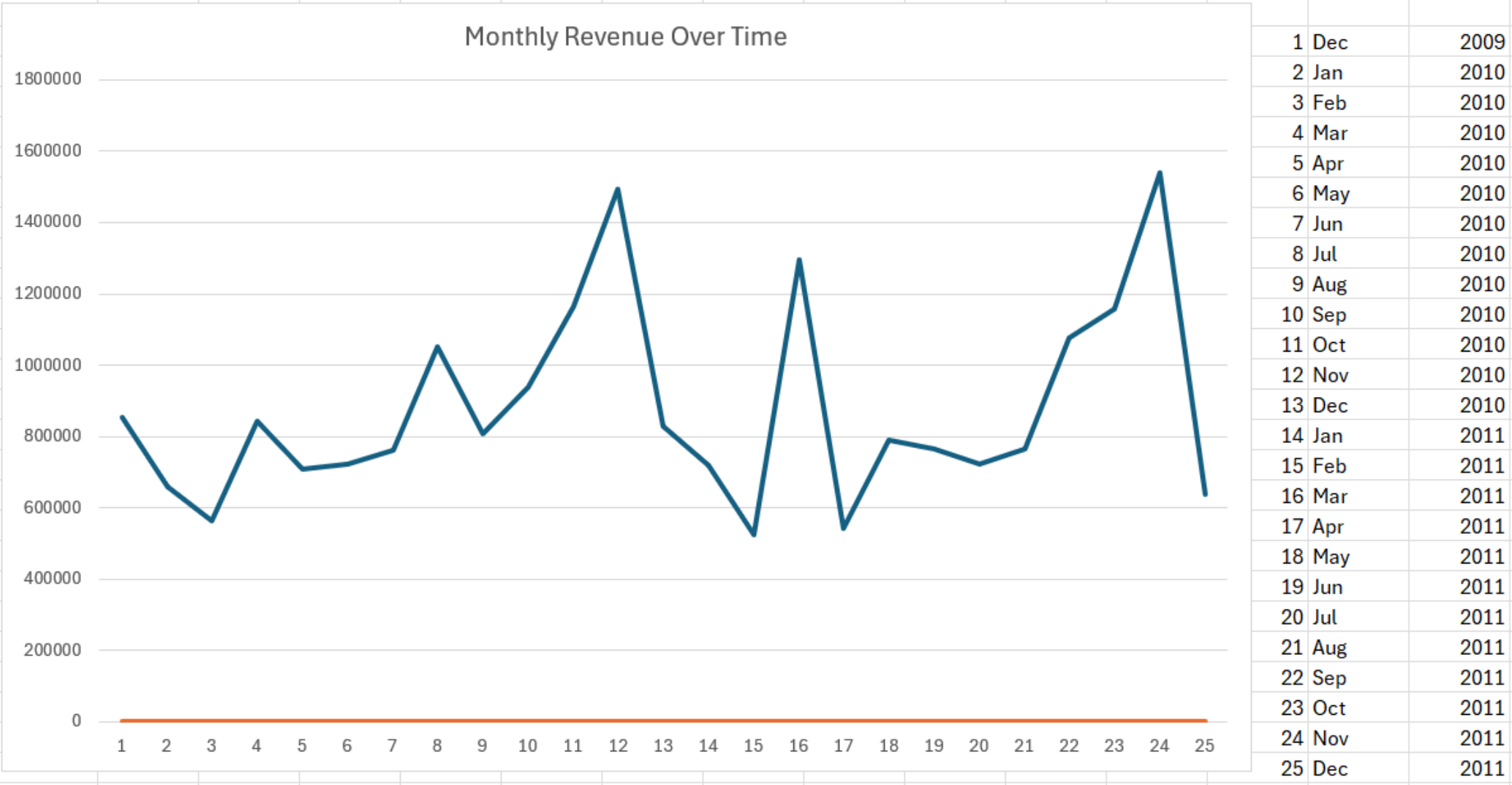
CUSTOMERS



RFM is a customer segmentation technique based on three factors:

- Recency (R): How recently the customer made a purchase.
- Frequency (F): How often the customer makes purchases.
- Monetary (M): How much money the customer spends.

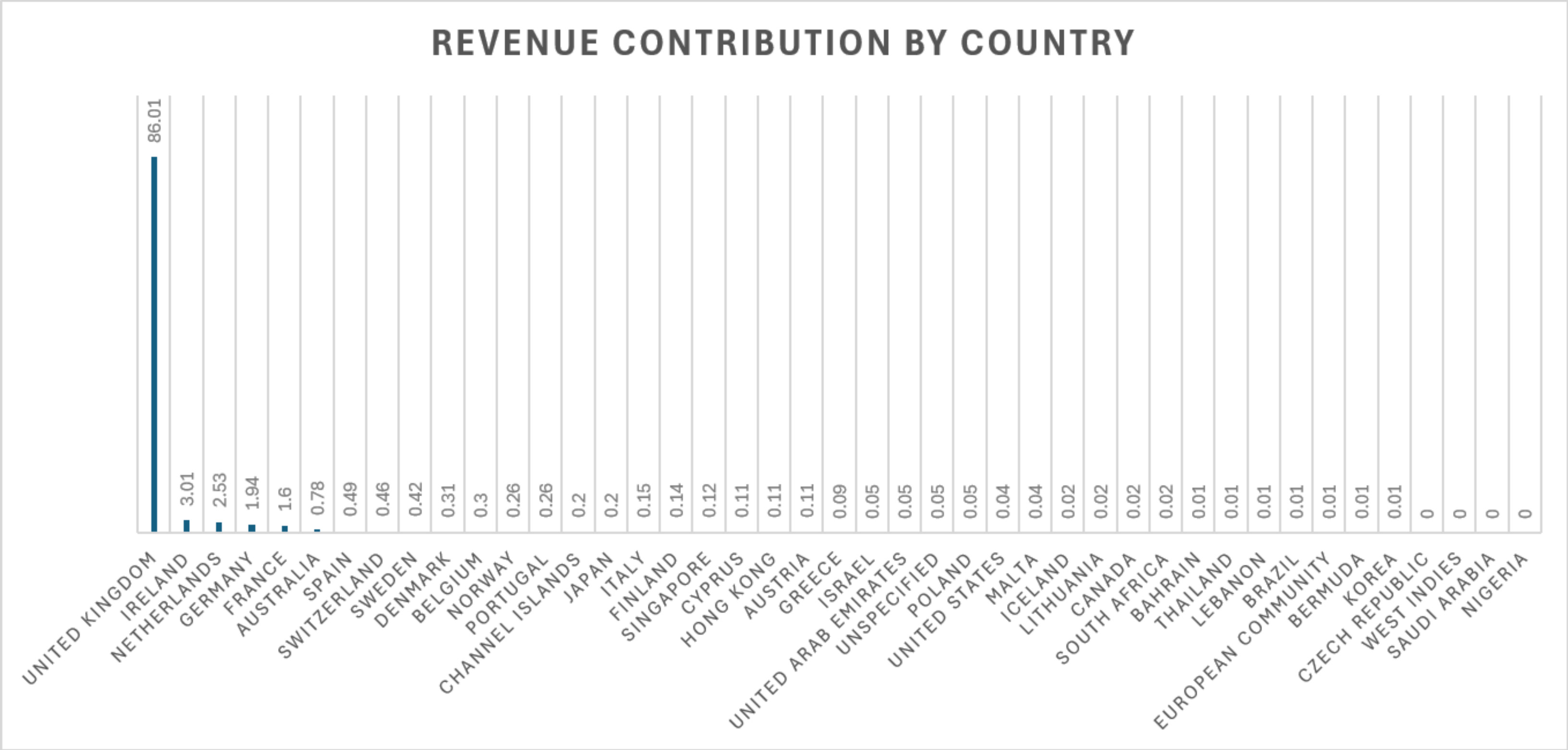
TIME



Monthly revenue trend over the two years

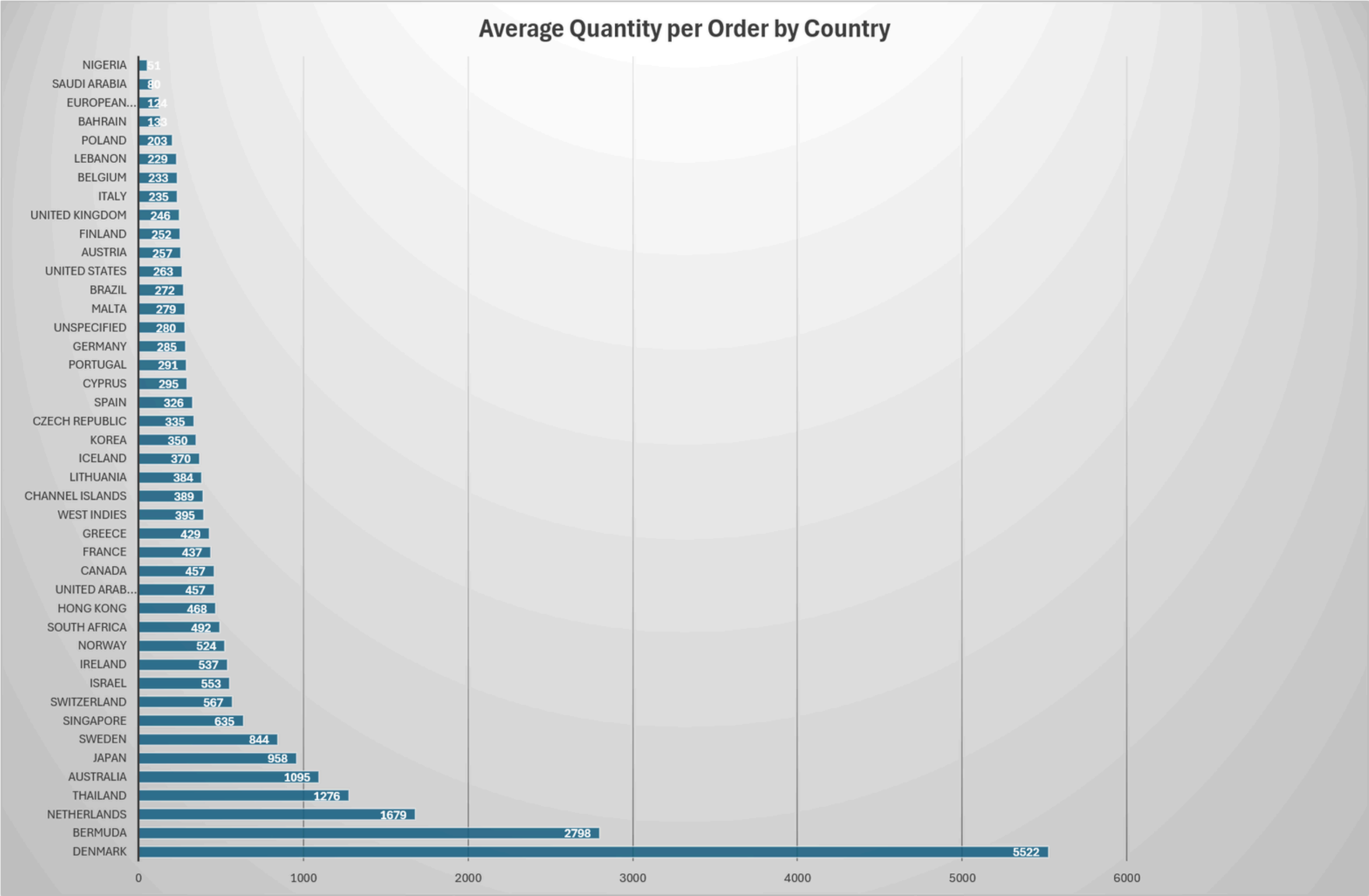
Revenue is higher during weekdays than weekends

COUNTRY



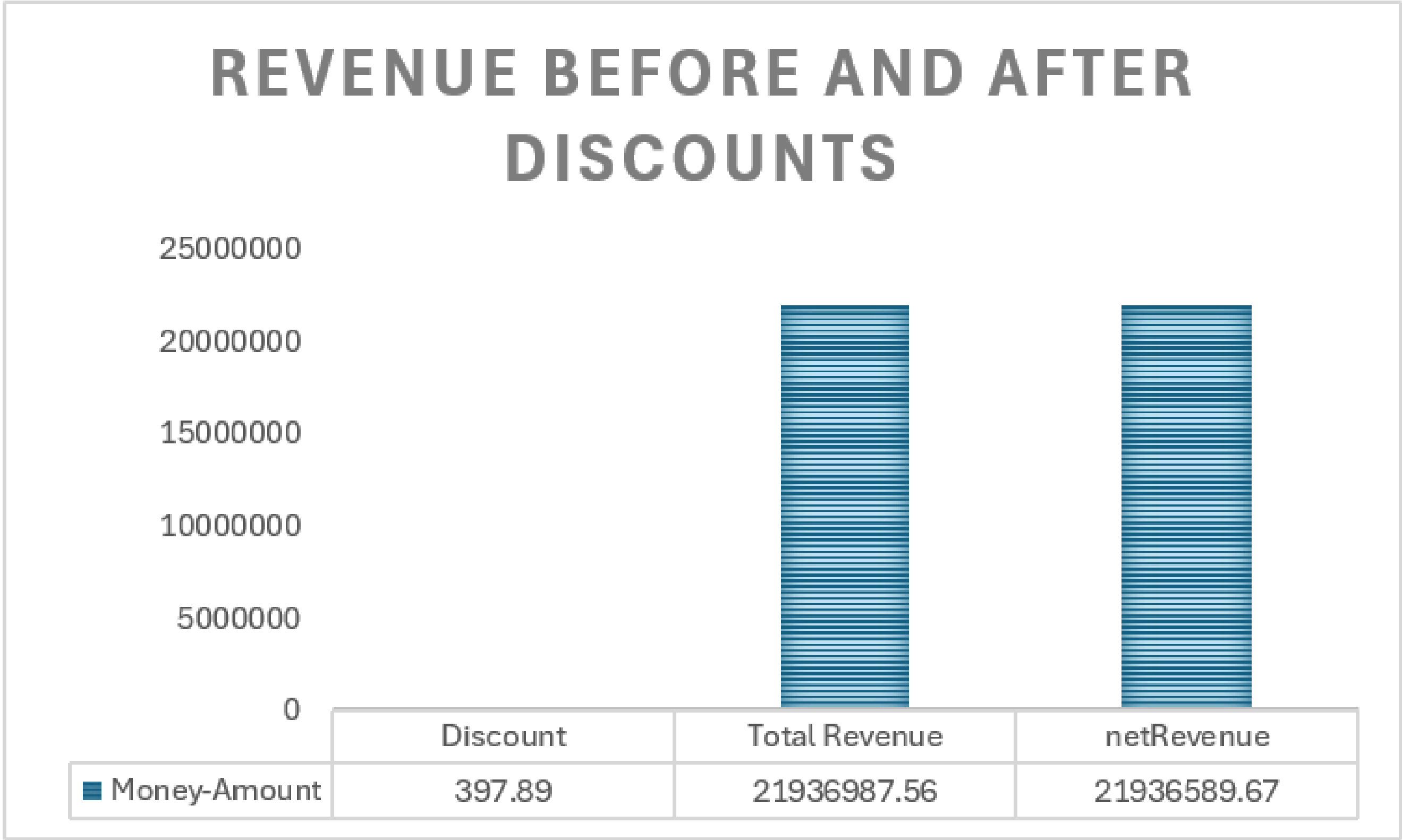
The United Kingdom contributed the highest share of total revenue

COUNTRY



Average order quantity varies significantly across different countries

NET REVENUE



Discounts had a very minimal impact on total revenue

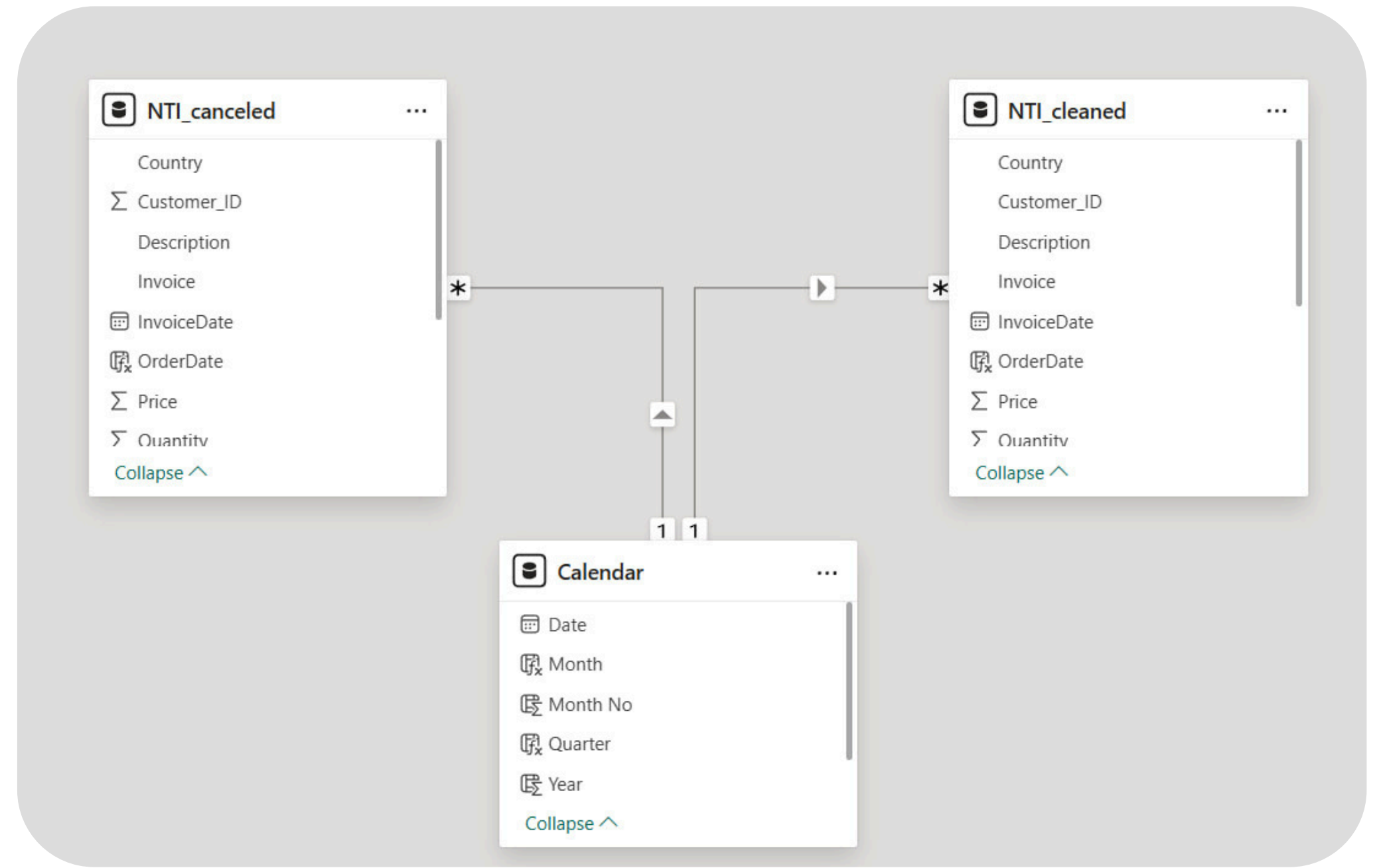
SALES & RETURNS DASHBOARD

Dashboard Objective:

- Analyze sales and returns performance.
- Track KPIs and business trends.
- Support data-driven decision making.

Dataset:

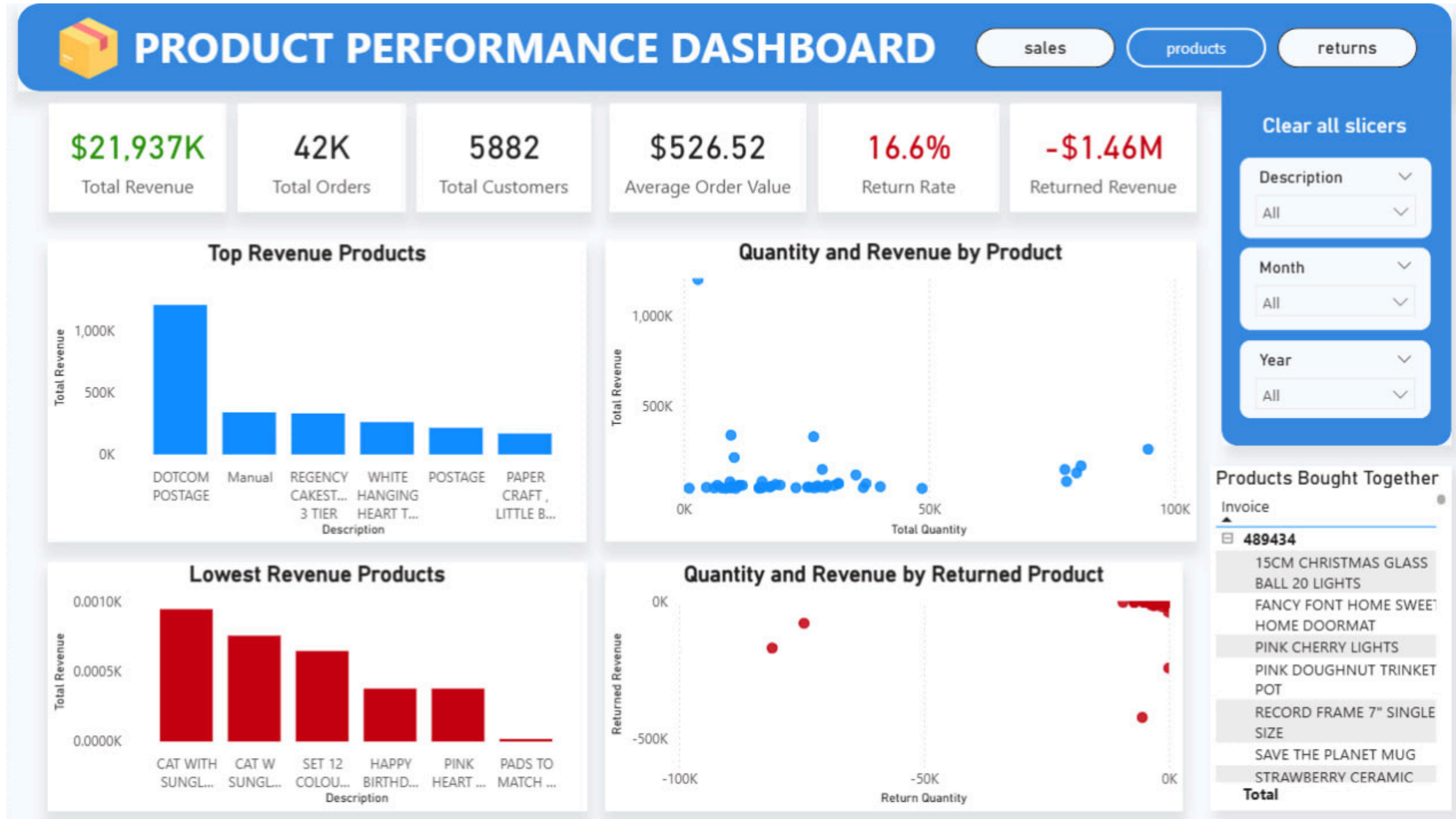
- Sales Data
- Returns Data
- Calendar Table (created on Power BI)



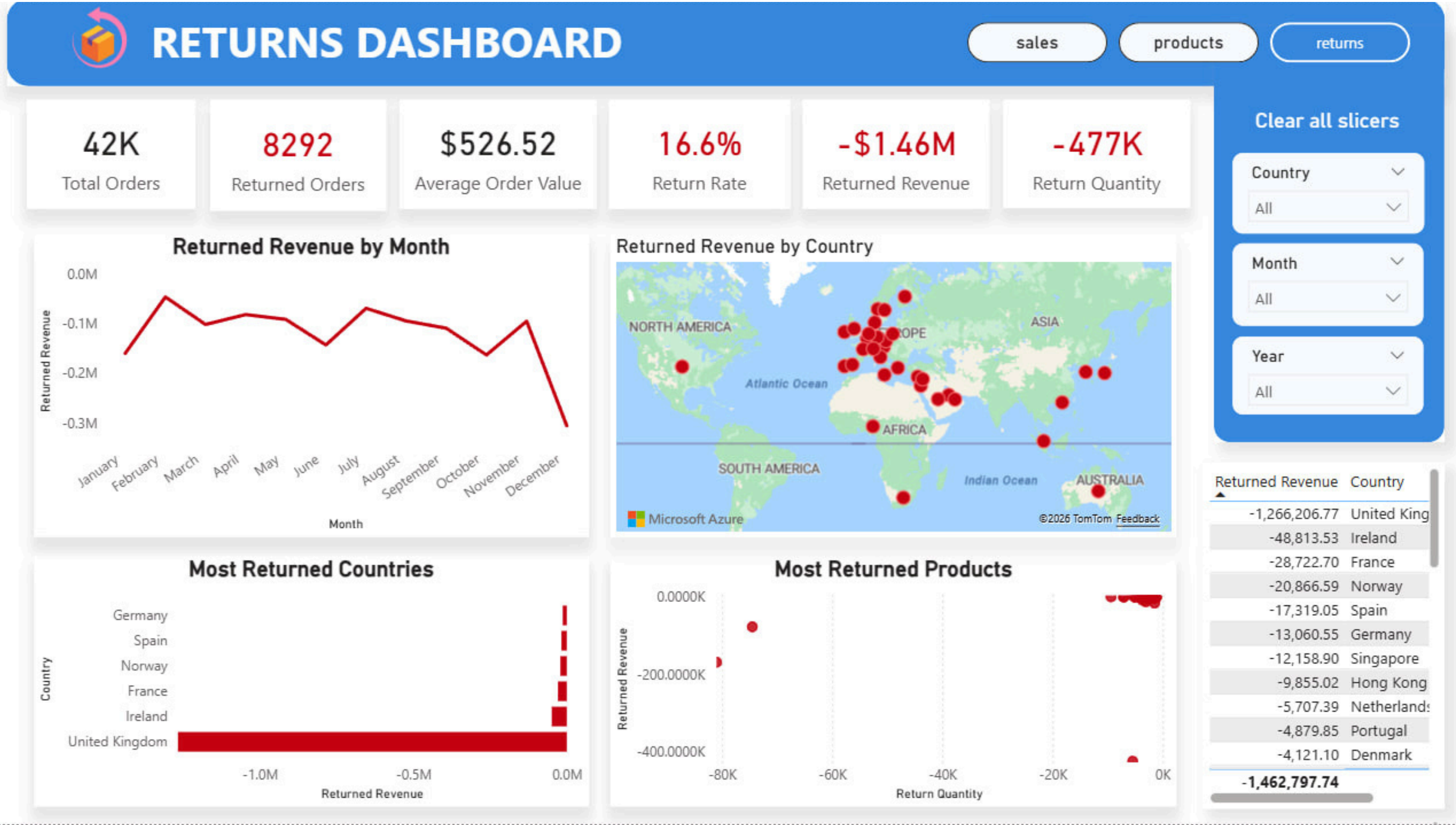
Page1: Sales & Returns Dashboard



Page2: Products Dashboard



Page3: Returns Dashboard



INSIGHTS

1. Product Performance

**Profitability is more important than sales volume.
High cancellation rates reduce revenue.**

2. Customer Behavior

**High-value customers drive most revenue.
Many customers are at risk of leaving.**

3. Country Performance

**The UK is the largest revenue contributor.
The business depends heavily on one market.**

4. Revenue Trend

**Sales vary by season and day.
Weekdays perform better than weekends.**

5. Product Association

**Customers frequently buy certain products together.
Bundling products can increase order value.**



RECOMMENDATIONS

1. Focus marketing on high-profit products.
2. Reduce product cancellations by improving quality and fulfillment.
3. Retain high-value customers and re-engage at-risk customers.
4. Increase marketing before peak seasons and launch weekend offers.
5. Expand into other countries to reduce dependence on the UK.
6. Create product bundles and cross-selling offers.



THANK
YOU

